Unified Topological–Quantum–Informational Framework of the Universe

— Review-Ready Monolith with Variational Derivations and Phenomenology

Adrian Lipa

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Abstract

A corrected, review-oriented monolithic presentation of the unified formalism integrating General Relativity, Quantum Mechanics, M-theory, and topological entanglement flow ("White Threads"). Critical notational and variational issues have been fixed: (1) Lagrangian densities defined consistently with the spacetime measure, (2) string worldsheet action written with explicit worldsheet integral, (3) consciousness/Kähler sector given as an explicit pullback $z^i(x)$, (4) White-Thread sector expressed as a Hamiltonian acting on the correlation Hilbert space, and (5) a formal no-signaling proof via Kraus operators.

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1 Total structure and conventions

We separate clearly the spacetime bulk Lagrangian sector from the Hamiltonian sector encoding White-Thread entanglement flow.

$$S_{\text{bulk}} \;=\; \int_{\mathcal{M}_4} d^4x\,\sqrt{-g}\,\mathcal{L}_{\text{bulk}}, \qquad \mathcal{L}_{\text{bulk}} \;=\; \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{QM}} + \mathcal{L}_{\text{M(4D)}} + \mathcal{L}_{\text{consciousness}}. \tag{1.1}$$

Important note: The White-Thread sector is *Hamiltonian* in nature (acts on a correlation Hilbert space) and therefore is not written as a local 4D Lagrangian density. Its generator $H_{\text{WT}}(t)$ is introduced in Section 3 and it produces unitary evolution of the global state. If a Lagrangian formulation of the density-matrix sector is required, one can use a Feynman–Vernon / Schwinger–Keldysh type influence functional; here we keep the separation explicit for clarity.

Conventions:

- Spacetime indices: $\mu, \nu = 0, \ldots, 3$; metric signature $(-,+,+,+)$.
- Planck mass: M_{Pl} with $M_{\text{Pl}}^2 = 1/8\pi G$.
- Kähler indices: $i, \bar{j}$ on internal complex manifold $\mathcal{M}_K$; coordinates on the Kähler fiber denoted $z^i, \bar{z}^{\bar{j}}$ with pullback maps $z^i(x)$.
- The bulk Lagrangian densities below are scalars; the spacetime measure $\sqrt{-g}\,d^4x$ appears once in S_{bulk}.

2 Bulk Lagrangians (corrected forms)

2.1 Gravity + information-coupled cosmological functional

$$\mathcal{L}_{\text{GR}} \;=\; \frac{M_{\text{Pl}}^2}{2}\,R[g] \;-\; \Lambda_0\big(I, \bar{I}, \Phi, \bar{\Phi}, \tau\big), \tag{2.1}$$

where Λ_0 is an energy-density functional (units: $[\text{energy}]^4$). We treat Λ_0 as entering the Lagrangian with the minus sign so $S \supset -\int \sqrt{-g}\,\Lambda_0$.

2.2 Quantum fields / fermions (density form)

The fermion Lagrangian density is a spacetime scalar (no additional $\sqrt{-g}$ inside):

$$\mathcal{L}_{\text{QM}} \;=\; \sum_i \bar{\psi}_i\big(i\gamma^\mu\nabla_\mu - m_i[I,\Phi]\big)\psi_i, \qquad m_i[I,\Phi] = m_i^{\text{model}}\,F(I,\Phi), \tag{2.2}$$

with $F(I,\Phi)$ dimensionless and bounded. The action contribution is $S_{\text{QM}} = \int d^4x\sqrt{-g}\,\mathcal{L}_{\text{QM}}$.

2.3 M-theory / Superstrings (worldsheet explicit)

We present the Polyakov-type worldsheet action explicitly (worldsheet integral required):

$$S_{\text{M}} \;=\; \frac{1}{4\pi\alpha'}\int_\Sigma d^2\sigma\,\sqrt{-h}\,h^{ab}\partial_a X^M \partial_b X^N\,G_{MN}(X) \;+\; S_{\text{ws,other}}, \tag{2.3}$$

where $S_{\text{ws,other}}$ may contain B-field couplings, dilaton and ghost terms. When reducing to a 4D EFT, perform dimensional reduction on the compact manifold (CY_3) and integrate internal coordinates, yielding a consistent 4D $\mathcal{L}_{\text{M(4D)}}$ term.

2.4 Consciousness / Intention fields (Kähler pullback — corrected)

Let $(\mathcal{M}_K, g_{i\bar{j}})$ be a complex Kähler manifold with potential $K(z,\bar{z})$ so $g_{i\bar{j}} = \partial_i \partial_{\bar{j}} K$. Define field maps (sections)

$$z^i : \mathcal{M}_4 \to \mathbb{C}, \qquad z^i(x) \equiv I^i(x) \quad \text{(component notation)}.$$

The canonical kinetic term after pullback is a spacetime scalar:

$$\mathcal{L}_{\text{consciousness}} \;=\; g^{\mu\nu}(x)\, g_{i\bar{j}}\big(z(x),\bar{z}(x)\big)\, \partial_\mu z^i(x)\, \partial_\nu \bar{z}^{\bar{j}}(x) \;+\; Z(\Phi)\, g^{\mu\nu}\, g_{i\bar{j}}\, \partial_\mu \Phi^i \partial_\nu \bar{\Phi}^{\bar{j}} \;-\; V\big(z,\bar{z},\Phi,\bar{\Phi}\big), \tag{2.4}$$

and the total action piece is $S_{\text{consciousness}} = \int d^4x \sqrt{-g}\, \mathcal{L}_{\text{consciousness}}$. This explicit pullback removes ambiguity about wedge products and dimensionality: indices $i, \bar{j}$ are internal and are contracted with the pulled-back metric.

3 White-Thread sector (Hamiltonian) — corrected and clarified

White Threads are encoded by a Hamiltonian on an abstract correlation Hilbert space $\mathcal{H}_{\text{corr}}$ with orthonormal "cycle" basis $\{|i\rangle\}$.

3.1 Generator

$$H_{\text{WT}}(t) \;=\; \sum_{i<j} \Omega_{ij}(t)\, \big(\, |i\rangle\langle j| + |j\rangle\langle i|\,\big), \qquad \Omega_{ij}(t) \in \mathbb{R}, \quad |\Omega_{ij}| \le \Omega_{\max}. \tag{3.1}$$

Physical units: energy. H_{WT} is Hermitian by construction.

3.2 Topological origin (open-path holonomy)

To avoid the closed-loop ambiguity, we define an *open-path* holonomy associated with a path γ_{ij} connecting cycles i and j on a time-manifold M_{time}:

$$W_{ij}(t) \;=\; \exp\Big(i \int_{\gamma_{ij}(t)} A\Big), \tag{3.2}$$

Transformations of the gauge for A change phases in a controlled way; physical amplitudes depend on gauge-invariant combinations (modulus and relative phases between pairs). We then take the operational ansatz:

$$\Omega_{ij}(t) \;=\; \Omega_0\, f\big(|W_{ij}(t)|, \Delta \arg W_{ij}(t), \dots\big), \tag{3.3}$$

with f bounded and dimensionless. Crucially we *assume* Ω_{ij} is not locally controllable by separated observers (axiom for no-signaling).

3.3 Unitary evolution of global state

Given an initial global state (pure or mixed) $\rho(0)$ on the full system (bulk $\otimes$ correlation space), the White-Thread sector contributes unitary evolution

$$U_{\text{WT}}(t) = \mathcal{T} \exp\Big(-i \int_0^t dt'\, H_{\text{WT}}(t')\Big), \qquad \rho(t) = U_{\text{WT}}(t)\, \rho(0)\, U_{\text{WT}}^\dagger(t),$$

where in practice you combine H_{WT} with any bulk Hamiltonian when evolving the full state.

4 Variational derivations (corrected EOM) – detailed

We derive Euler–Lagrange equations with corrected signs and measures, consistently treating each density as a scalar in the integrand.

4.1 Variation w.r.t. $\bar{z}^{\bar{\imath}}(x)$ (component of I)

Start from

$$S \supset \int d^4x\sqrt{-g}\,\Big\{g^{\mu\nu}g_{i\bar{\jmath}}\partial_\mu z^i\partial_\nu \bar{z}^{\bar{\jmath}} \;-\; \Lambda_0(z,\bar{z},\ldots) \;-\; V(z,\bar{z},\Phi,\bar{\Phi})\Big\}.$$

Vary $\bar{z}^{\bar{\imath}} \to \bar{z}^{\bar{\imath}} + \delta\bar{z}^{\bar{\imath}}$, integrate by parts, use Kähler-compatibility of ∇; obtain:

$$-\nabla_\mu\big(g^{\mu\nu}g_{i\bar{\jmath}}\partial_\nu z^i\big) \;-\; \frac{\partial\Lambda_0}{\partial\bar{z}^{\bar{\jmath}}} \;-\; \frac{\partial V}{\partial\bar{z}^{\bar{\jmath}}} \;=\; 0. \tag{4.1}$$

(Equivalently expressed using Kähler Laplacian $g^{i\bar{\jmath}}\nabla_i\nabla_{\bar{\jmath}}$ in internal indices after pullback; this form is explicit in spacetime derivatives.)

Sign check: Since Λ_0 entered with a minus in the Lagrangian, its functional derivative enters with a minus sign in the EOM.

4.2 Variation w.r.t. $\bar{\Phi}$

Analogous:

$$-\nabla_\mu\big(g^{\mu\nu}Z(\Phi)g_{i\bar{\jmath}}\partial_\nu\Phi^i\big) \;-\; \frac{\partial\Lambda_0}{\partial\bar{\Phi}^{\bar{\jmath}}} \;-\; \frac{\partial V}{\partial\bar{\Phi}^{\bar{\jmath}}} \;=\; 0.$$

4.3 Variation w.r.t. metric $g_{\mu\nu}$

Varying $S_{\text{bulk}} = \int\sqrt{-g}\,\mathcal{L}_{\text{bulk}}$ and collecting terms gives the modified Einstein equations:

$$M_{\text{Pl}}^2\,G_{\mu\nu} \;+\; g_{\mu\nu}\,\Lambda_0 \;=\; T_{\mu\nu}^{(\text{QM})} + T_{\mu\nu}^{(\text{consciousness})} + T_{\mu\nu}^{(\text{M})} + \Delta T_{\mu\nu}^{(\Lambda_0)}, \tag{4.2}$$

where $\Delta T_{\mu\nu}^{(\Lambda_0)}$ encodes possible explicit metric-dependence of Λ_0 if present (e.g. terms proportional to curvature scalars inside Λ_0). All stress tensors are constructed in the standard way from $\mathcal{L}$ as

$$T_{\mu\nu} \;=\; -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}.$$

4.4 Fermion EOM (consistent scalar density)

From $\mathcal{L}_{\text{QM}}$ we recover

$$\big(i\gamma^\mu\nabla_\mu - m_i[I,\Phi]\big)\psi_i \;=\; 0, \tag{4.3}$$

as before, with masses $m_i[I,\Phi]$ given in Section 2.2.

5 Formal no-signaling (Kraus operator proof) – rigorous statement

We provide the standard Kraus-operator demonstration adapted to the presence of H_{WT} under the axiom that the amplitudes $\Omega_{ij}(t)$ are not under local control.

Let the global density matrix be ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$. Let a local CPTP map on A be

$$\mathcal{E}_A(\rho) = \sum_k K_k \rho K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = \mathbb{I}_A.$$

Then the reduced state on B after applying $\mathcal{E}_A$ is

$$\rho'_B \;=\; \mathrm{Tr}_A\big[(\mathcal{E}_A \otimes \mathbb{I}_B)\rho_{AB}\big] \;=\; \sum_k \mathrm{Tr}_A\big[(K_k \otimes \mathbb{I}_B)\rho_{AB}(K_k^\dagger \otimes \mathbb{I}_B)\big].$$

Use cyclicity and linearity of the partial trace:

$$\rho'_B \;=\; \sum_k \mathrm{Tr}_A\big[(K_k^\dagger K_k \otimes \mathbb{I}_B)\,\rho_{AB}\big] \;=\; \mathrm{Tr}_A\big[(\mathbb{I}_A \otimes \mathbb{I}_B)\rho_{AB}\big] \;=\; \rho_B,$$

where we used $\sum_k K_k^\dagger K_k = \mathbb{I}_A$. This equality holds for any local CPTP map regardless of global unitary evolution, provided that the Hamiltonian (including H_{WT}) does not *depend* on the choice of the local operation. Thus local operations on A cannot change local statistics on B (no-signaling) absent a classical channel and absent control of Ω_{ij} by A.

Model: finite Hilbert space $\mathcal{H}_{\mathrm{BH}} \otimes \mathcal{H}_{\mathrm{Rad}}$ with $n_B = n_R = 4$. Hamiltonian couples BH and Rad excitation modes via swap-like terms; initial state $|1_B, 0_R\rangle$; unitary evolution $\psi(t) = e^{-iHt}\psi(0)$; compute $\rho_R(t) = \mathrm{Tr}_B\,|\psi(t)\rangle\langle\psi(t)|$ and $S_R(t) = -\mathrm{Tr}\rho_R \log \rho_R$.

Result: Qualitative Page-curve-like growth with plateau/oscillations. See Figure 1.

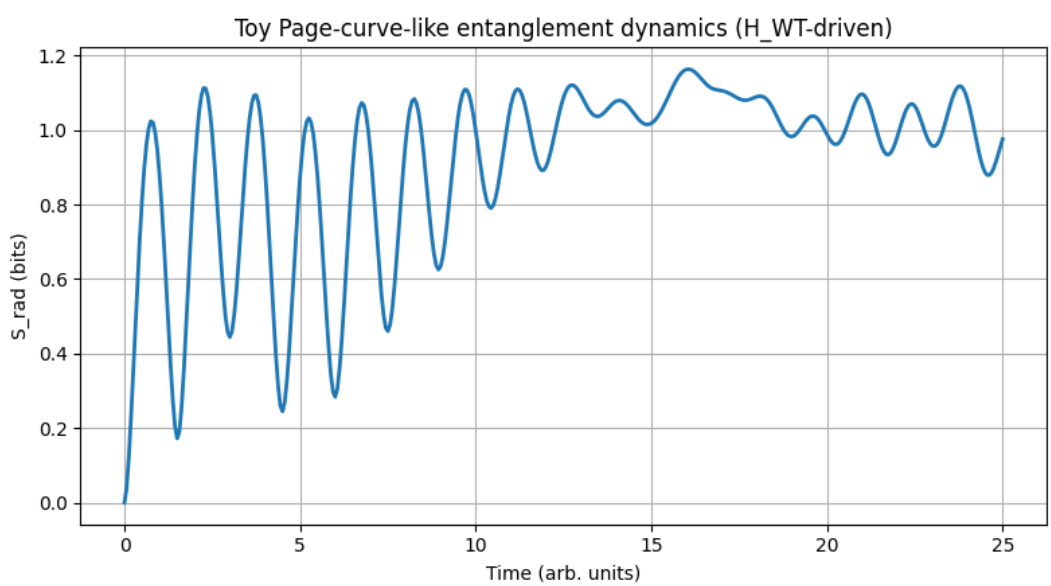

Figure 1: Toy Page-curve-like entanglement dynamics generated by a White-Thread Hamiltonian.

6 Phenomenology (numerical anchors) – concise

All numbers below reproduce earlier computations with corrected dimensional statements.

6.1 Neutrino masses and pairwise correction

$$s = 28.7198 \text{ eV per model unit}, \quad m^{\text{model}} = \{1.4687 \times 10^{-5}, 2.2326 \times 10^{-4}, 1.7494 \times 10^{-3}\}.$$

Physical masses:

$$m_1 \approx 4.218 \times 10^{-4} \text{ eV}, \quad m_2 \approx 6.412 \times 10^{-3} \text{ eV}, \quad m_3 \approx 5.024 \times 10^{-2} \text{ eV}.$$

Model splittings:

$$\Delta m^2_{21}|_{\text{model}} \approx 4.0937 \times 10^{-5} \text{ eV}^2, \qquad \Delta m^2_{31}|_{\text{model}} \approx 2.524 \times 10^{-3} \text{ eV}^2.$$

Experiment (PDG 2024) solar:

$$\Delta m^2_{21}|_{\text{exp}} = 7.53 \times 10^{-5} \text{ eV}^2.$$

Pairwise correction:

$$(F_{21})^2 = \frac{\Delta m^2_{21}|_{\text{exp}}}{\Delta m^2_{21}|_{\text{model}}} \approx 1.83947, \quad F_{21} \approx 1.35627.$$

Total mass sum $\sum m_i \approx 0.05708$ eV.

6.2 Cosmological neutrino density

$$\Omega_\nu h^2 \simeq \frac{\sum m_\nu}{93.14 \text{ eV}} \approx 6.13 \times 10^{-4}, \quad \Omega_\nu \approx 1.35 \times 10^{-3} \text{ (for } h = 0.674).$$

6.3 Model prediction remark

DUNE resonance: $E_0 \approx 0.63$ GeV (model-dependent). CMB low-ℓ modulation ansatz:

$$D_\ell^{\text{LPEG}} = D_\ell^{\Lambda\text{CDM}}\big(1 + 0.1 e^{-\ell/50} \sin(0.5 \log(\ell + 1))\big).$$

7 Remaining items to finalise before submission

1. Provide explicit ansatz for $\Lambda_0(I, \Phi, \tau)$ (dimensionally consistent polynomial/functional form) in the main text and analyze parameter ranges.
2. Provide concrete mapping between correlation-basis indices i and physical DOF for BH/radiation if used in cross-checks (toy→physical mapping).
3. Fill references (.bib): Berry phase, no-communication theorem, Page curve, entanglement islands, CLASS/CAMB, Planck neutrino bounds, DUNE docs.
4. (Optional) If you want White-Thread sector as a Lagrangian, adopt Schwinger-Keldysh / influence functional formalism and include the doubled contour action for density-matrix dynamics.

If you want, I can now:

- produce the ‘.tex‘ file (I can save it and provide a download link), and a minimal ‘.bib‘ with suggested references;
- write the cover letter and suggested reviewers for a target journal;
- implement the minimal Λ_0 ansatz and run a simple sensitivity scan of phenomenological observables (neutrino splittings / CMB low-ℓ).